

Frontend Implementation

The frontend of the **DarshanEase – Temple Darshan Ticket Booking System** is developed using **React.js** and **Vite**. It provides an interactive, responsive, and user-friendly interface that allows devotees to browse temples, book darshan tickets, make payments, and manage their profiles.

1. React Application Setup

The frontend application is initialized using **React.js** with **Vite**, providing a fast development environment and optimized build process.

The frontend setup includes the following tasks:

- Create the React application using Vite.
- Configure project structure with separate folders for components, pages, services, routes, context, hooks, assets, and utilities.
- Configure client-side routing using **React Router DOM**.
- Install the required libraries and dependencies such as:
 - React Router DOM
 - Axios
 - React Icons
 - React Toastify
 - Bootstrap or CSS
 - JWT Decode (if required)

These libraries provide navigation, API communication, notifications, styling, and authentication support.

2. User Interface Design

The **DarshanEase** application provides an intuitive and responsive user interface for devotees, organizers, and administrators.

The UI development includes:

Create Reusable Components

Develop reusable React components such as:

- Navbar
- Footer
- Temple Card
- Booking Form
- Time Slot Card
- Search Bar
- Button
- Loader
- Payment Form

These reusable components improve code maintainability and reduce duplication.

Implement Layout and Styling

The application layout is designed using modern CSS techniques to ensure a responsive experience across desktops, tablets, and mobile devices.

The interface includes:

- Home Page
- Login Page
- Registration Page
- Temple Listing Page
- Temple Details Page
- Darshan Slot Selection
- Ticket Booking Page
- Payment Page
- Booking Confirmation Page
- My Bookings Page
- User Profile Page
- Admin Dashboard

Proper styling is applied using CSS to provide a clean, attractive, and user-friendly experience.

Navigation

Navigation between pages is implemented using **React Router DOM**.

The application supports smooth navigation between:

- Home
- Login
- Register
- Temples
- Book Ticket
- Payment
- My Bookings
- Profile

- About
- Contact
- Admin Dashboard

Protected routes are implemented to restrict access to authenticated users only.

3. Frontend Logic Implementation

The frontend communicates with the backend using RESTful APIs to provide dynamic functionality.

API Integration

Axios is used to connect the frontend with backend API endpoints.

API integration is implemented for:

- User Registration
- User Login
- Fetch Temple Details
- Fetch Darshan Slots
- Book Darshan Ticket
- Payment Processing
- View Booking History
- Update User Profile
- Admin Operations

The frontend sends HTTP requests to the backend and displays the received data dynamically.

Data Binding

React state management is used to bind data between the user interface and backend services.

Dynamic data binding is implemented for:

- User Information
- Temple Listings
- Darshan Slot Availability
- Booking Details
- Payment Status
- Booking History
- Profile Information

Whenever data changes in the backend, the frontend updates automatically without requiring a page refresh, providing a smooth user experience.